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Engineering Sonogenetic EchoBack-CAR T cells

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Summary

Chimeric antigen receptor (CAR) T cell therapy for solid tumors encounters challenges such as on-target off-tumor toxicity, exhaustion, and limited T cell persistence. Here, we engineered sonogenetic EchoBack-CAR T cells using an ultrasensitive heat-shock promoter screened from a library, integrated with a positive feedback loop from CAR signaling, enabling long-lasting CAR expression upon focused-ultrasound (FUS) stimulation. EchoBack-hGD2CAR T cells, targeting disialoganglioside GD2, exhibited potent cytotoxicity and persistence in 3D glioblastoma (GBM)

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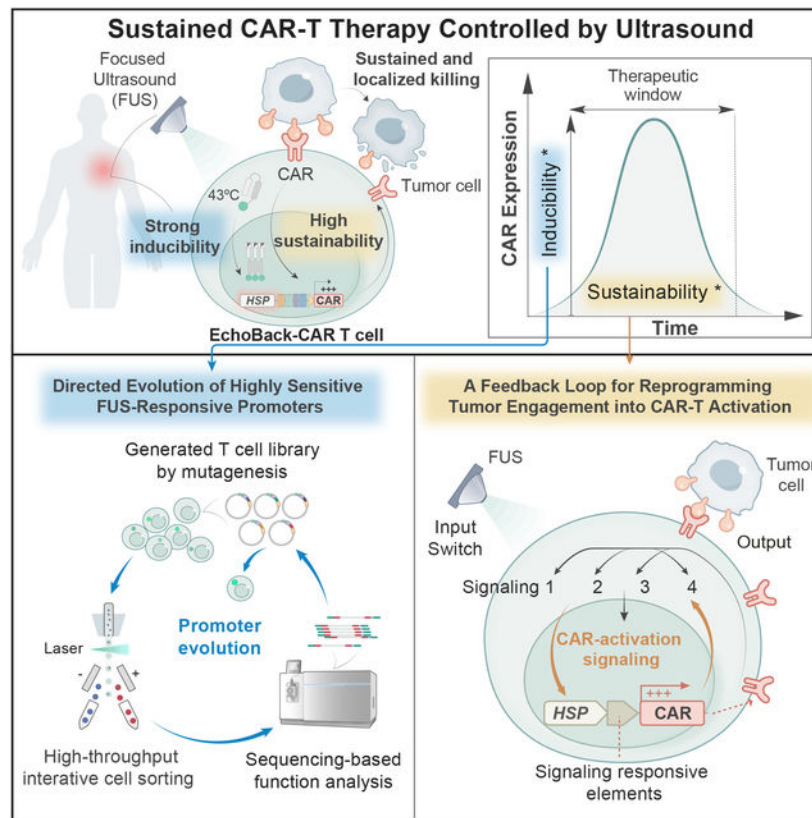
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Declaration of interests

Yingxiao Wang is scientific co-founder and consultant of Cell E&G Inc. and Acoustic Cell Therapy Inc. Dulei Li is an employee of Acoustic Cell Therapy Inc. A patent related to this study has been filed. These financial interests do not affect the design, conduct or reporting of this research.

models. In mice, EchoBack-hGD2CAR T cells suppressed GBM without off-tumor toxicity and outperformed their constitutive counterparts. Single-cell RNA sequencing revealed enhanced cytotoxicity and reduced exhaustion in EchoBack-CAR T cells, compared to the standard CAR T cells. This EchoBack design was further adapted to target prostate-specific membrane antigen (EchoBack-PSMACAR) for prostate cancer treatment, demonstrating long-lasting tumor suppression with minimal off-tumor toxicity. Thus, the Sonogenetic EchoBack-CAR T cells can serve as a versatile, efficient, and safe strategy for solid tumor treatment.

Graphical Abstract



IN BRIEF

Sonogenetic EchoBack-CAR T cells were designed with an ultrasensitive heat-shock promoter screened from a library and a synthetic positive feedback loop that reprograms tumor engagement into CAR-T cell activation. In this approach, a short burst of focused ultrasound stimulation triggers long-lasting CAR-T cell activation and continuous cancer cell destruction while maintaining minimal toxicity.

Keywords

Sonogenetics; CAR-T; Promoter Library Screening; Feedback Engineering; Solid Tumor; Immunotherapy

Introduction

Immunotherapy based on CAR-expressing T cells has been shown to cure leukemia and other cancers in clinical trials.^{1–3} Despite clinical successes in treating blood cancers, major challenges remain before CAR-based immunotherapy can be adopted for treating solid tumors, such as the life-threatening non-specific targeting of the CAR T cells against normal/nonmalignant tissues (on-target off-tumor toxicities; OTOT),^{1,4–11} T cell exhaustion^{12–14} and poor T cell persistence after infusion into patients.¹⁵ In fact, off-tumor toxicities against the lung, gray matter in the brain, and cardiac muscles have caused multiple cases of death.^{16–18} Recent advancements in synthetic biology and controllable genetics,^{19–22} such as sonogenetics, allow for remote-controlled gene activation with high spatiotemporal precision, mitigating off-tumor toxicity in CAR T cell therapies.^{20,23} These technologies enable the targeting of antigens with high efficiency but suboptimal specificity, alleviating antigen selection constraints for solid tumor treatment; however, the limited inducibility and short lifespan of CAR expression in these systems reduce their overall efficacy. Thus, there is an urgent need for a platform to identify potent inducible promoters and strategies to enhance the sustainability of CAR expression for practical therapeutic efficacy and clinical application.

Promoters are regulatory DNA regions controlling transcription initiation, determining the induction potency of inducible systems. While saturation point mutagenesis has been used to optimize promoters,²⁴ including heat shock elements (HSE) controlled by heat shock, better promoters cannot be identified by point mutation alone due to the greater-than-additive (synergistic) effect of multiple activator-binding sites. Thus, a systematic approach is needed for promoter optimization. In addition, the rapid downmodulation of CAR molecules on the T cell surface in inducible systems impairs CAR-T functionality,^{15,25} which needs to be mitigated. Upon recognition and engagement with tumor cells, CAR-T cells activate various interdependent signaling pathways that coordinate immune responses,²⁶ which include the calcium signaling,²⁷ ZAP70 kinase pathway,²⁸ mitogen-activated protein kinases (MAPKs) and extracellular signal-regulated kinases (ERKs) pathway,^{27,28} nuclear factor of activated T-cells signaling (NFAT),²⁹ nuclear factor kappa B signaling,²⁹ and cyclic AMP (cAMP) pathway.³⁰ These pathways collectively enable CAR-T cells to effectively target and kill cancer cells. We propose to investigate whether tumor engagement-induced signaling can be harnessed and redirected to drive more sustained CAR production with a transient stimulation by ultrasound, thereby enhancing the persistence of CAR function in T cells.

Results

The design and optimization of sonogenetic EchoBack-CAR

In this study, we engineered EchoBack-CAR T cells, a sonogenetically controllable CAR T cell, utilizing a highly heat-sensitive promoter evolved through a sort-seq platform²⁸ and integrated it with a positive feedback genetic circuit that can convert tumor cell engagement and the subsequent activation of CAR signaling pathways into CAR production (Figure 1A). This EchoBack design enables sustained CAR expression and tumor elimination upon remote and noninvasive stimulation by a transient short pulse of focused ultrasound, offering

a practical solution to the challenges of OTOT toxicity, exhaustion, and limited persistence of therapeutic T cells in solid tumors.

Screen and evolve heat-sensitive promoter via sort-seq

The heat shock promoters, such as the HSP70 promoter, have been used to induce CAR in T cells for immune therapy.^{23,31} To improve the function of heat shock promoter, saturation point mutagenesis of each individual residue was performed separately on the heat shock element (HSE) of the promoter DNA sequence, and an effective unit sequence of HSE was identified by combining the best mutation of different residues.^{24,32} However, this simple combination may not lead to the optimal HSE performance, as one HSE binding site may affect the binding of others to heat shock transcription factor (HSF), a regulatory transcription factor of HSE.

To increase the inducibility of HSE, we have developed a sort-seq platform to evolve and identify potent inducible promoters, leveraging a global site-saturation mutagenesis, T cell library screening and sorting, and next generation sequencing (Sort-seq, Figure 1B). We first integrated a single HSE derived from the human Hsp 70.1 promoter, and a synthetic YB-TATA minimal promoter to drive the expression of EGFP as the screening reporter, with constitutively expressed mCherry serving as an internal expression reference. After a global site-saturated mutagenesis (Figure S1A and S1B, nGAAnnTTCnnGAnn, in which n represents any nucleotide type), a library containing 16,384 different HSE variants that covers every possible combination were generated and introduced into mammalian cells (Figure 1B),²⁸ with the mutagenesis validated by sanger sequencing (Figure S1C). We then chose the T cell line for library generation and screening and controlled the lentivirus multiplicity of infection (MOI) to be at 0.1 so that ~95% of the infected T cells receive only one kind of HSE variants in a single cell. To screen for better HSE candidates, we performed iterative counter sorting of higher induction (EGFP high) and lower leakage (EGFP negative) using high-throughput fluorescence-activated cell sorting (FACS) (Figure S1D). This counter sorting strategy resulted in the selection of variants with higher induction rate after heat shock at 43 °C for 15 minutes while maintaining a low leakage expression at rest 37°C culture conditions. Consequently, this approach enabled the continual enrichment of the candidates with desired functions, as represented by their gradually increasing frequency after each round of screening (Figure 1C and 1D).

The input group of cells before cell sorting and the enriched cells after each round of selection were collected for next-generation sequencing, and normalized reads of each promoter variant (frequency) were quantified and ranked among all the variants. After three iterative rounds of high-induction enrichments and two rounds of low-leakage selection, the top candidates were enriched and identified from the library. Some clones within the library exhibited progressive enrichment, whereas others showed a gradual decay, illustrating the selectivity of enrichment effect (Figure 1C). The top 5 HSE candidates with the most significant enrichment in the NGS data were selected for further verification (Figure 1D), using the rationally designed HSE²⁴ and the parental native Hsp 70.1 HSE as control. EGFP was used as the testing reporter, driven by a single copy of the HSE variants. Most of the selected variants have improved induction rate compared to the parental or rationally

designed HSE, as represented by the higher induction rate and mean fluorescent intensity of EGFP (Figure 1E and S1E). Notably, among the top 5 ranked clones, clone #4 has significantly stronger induction of EGFP expression after heat shock, even when induced at a shorter heat shock duration, such as 43°C for 5 minutes or 45°C for 1.5 minutes (Figure 1E and S1E).

An optimized promoter enables stronger induction of CAR expression in T cells

We next investigated whether the optimized HSE can be used to effectively induce CAR expression in T cells (Figure 1F). We started with a high affinity anti-GD2 CAR (hGD2CAR) using an scFV derived from 14G2a carrying a E101K mutation,^{33,34} CD28 transmembrane and costimulatory domain, and CD3 ζ chain. For this hGD2CAR, controlled expression in a confined tumor region is necessary, as its continuous expression has been demonstrated to exhibit fatal toxicity when systematically applied *in vivo*.³³ We elected to advance with the identified promoter clone #4, which exhibited markedly enhanced induction, particularly in response to weaker and medically more compatible heat shock stimulations. We found a single element of HSE failed to induce a significant CAR expression, which is significantly bigger than EGFP and requires a membrane display on the T-cell surface (Figure S1F). We hence engineered seven repeats of the clone #4 sequence (7HH) to drive the expression of hGD2CAR. The 7HH displayed a significantly stronger induction of CAR expression in T cells than the seven repeats of rationally designed (RD-7H) or parental HSE (parent-7H) while maintaining a low level of leakage expression (less than 5%) (Figure 1F and 1G). This enhancement was particularly noticeable with short pulsed stimulations (Figure 1G and S1G), favorable in clinics as they can avoid potential tissue damage and be convenient for patients. Specifically, the promoter based on clone #4 HSE demonstrates a 4.32-fold and 5.88-fold increase in CAR induction compared to the one based on parent HSE under stimulations at 43°C 5 minutes and 45 °C 1.5 minutes, respectively. In summary, utilizing Sort-seq, we successfully performed the promoter library screening and identified a promoter that has significantly improved performance in response to heat shock stimulation comparing to the parental template.

Design a positive feedback loop to elongate the CAR expression

To strengthen the persistence of the inducible CAR in this controllable system, we investigated whether tumor engagement and its induced signaling can be harnessed to drive more persistent CAR production in response to transient stimulations. We developed a positive feedback loop to guide the induced and activated CAR signals into the synthesis of more CAR molecules, which can counteract the antigen-engagement-induced CAR downregulation (Figure 2A). We assembled and compared various combinations of minimal promoters, CARs, and regulatory elements (Figure 2B and S2A). The expression of inducible hGD2CAR without feedback (feedback design #1) decreased significantly in 24 hours after induction in monoculture. When co-cultured with GD2⁺ U-87 MG tumor cells, antigen engagement triggered further CAR reduction in T cells, leading to a complete CAR disappearance on surface within 24 hours (Figure 2C). When we added the NF κ B responsive elements into the promoter region to utilize the NF κ B signaling (feedback design #2), it increased the sustainability of CAR on T cell surfaces in monoculture but did not reverse the reduction of CAR on T cells when cocultured with tumor cells (Figure 2D). Since the

IL2 mini-promoter has been reported to promote the inducible gene expression,^{35,36} we examined it but found that the YB-TATA mini-promoter performed better both in feedback designs #1 and #2 (Figure S2B and S2C). We then added NFAT responsive elements into the promoter region containing the NF κ B element and YB-TATA promoter (feedback design #3), which reduced the downregulation of CAR when engaged by antigen (Figure 2E). It has been reported that 41BB can induce a sustained NF κ B activation,³⁷ but at the same time can trigger CAR endocytosis and degradation via the ubiquitination mechanism.¹⁵ We hence examined whether the incorporation of an additional 41BB costimulatory domain can enable long-lasting effects on CAR expression by activating NF κ B signaling as a net effect. However, the results showed that an additional 41BB costimulatory domain caused more CAR reduction upon antigen engagement (Figure S2D), which suggests that the 41BB-related endocytosis process is stronger than the effect on CAR production. Only when an additional CRE-SRE element responsive to the MAPK and cAMP signaling was incorporated into the promoter, together with NFAT-NF κ B and YB-TATA, for the expression control of CAR containing the CD28 but not 41BB costimulatory domain (feedback design #4), a high induction and sustainability can be achieved while maintaining a low level of basal leakage before heat shock stimulation (Figure 2F and S2G). In fact, after heat shock, this feedback design #4 reversed the downregulation trend of CAR, leading to the CAR surface expression for over three days after co-cultured with tumor cells, in contrast to that of less than one day for the parental feedback design #1 (Figure 2F and S2G).

Positive feedback integration enables robust tumor elimination *in vitro*

We then examined the killing efficacy of various sonogenetic hGD2CAR feedback designs. The inclusion of feedback elements resulted in a relatively low leakage, without impacting the inducibility of the heat shock promoter in primary human primary T cells (Figure S3A and S3B). The killing efficacy of these hGD2CAR variants after heat shock was further measured at different effector-to-target (E:T) ratios and the results indicate that the feedback elements significantly enhanced CAR functions. Our feedback design #4 resulted in the most significant killing of tumor cells even at a low E:T ratio of 1:4 and at different durations of co-cultures (Figure S3C and S3D).

Live cell analysis was further employed to verify and quantify the dynamics of inducible CAR-T cell proliferation and tumor cell lysis. Consistent with the Fluc assay, all these feedback design variants showed significant tumor killing after activation, in contrast to the non-engineered naïve T cells (Figure 2G); however, obvious tumor cell relapse was observed in the group treated by the parental design #1 after 48 hours, while the feedback design #4 achieved sustained tumor suppression (Figure 2G, 2H, and 2I). Consistently, this feedback design #4 also showed a sustained CAR T proliferation while parental design #1 showed a transient increase and quick decay during co-culture (Figure 2H). These results demonstrated the successful engineering of sonogenetic feedback CAR-T cells, which enables sustained tumor elimination *in vitro*.

EchoBack-hGD2CAR T cells exhibit potent cytotoxicity

We next integrated the optimized 7HH promoter (promoter clone #4, Figure 1) and positive feedback loop elements (feedback design #4, Figure 2) to drive the CAR expression,

which is named EchoBack-CAR (Figure 3A). We comprehensively evaluated the dose-response curve and on/off kinetics of the promoter, which showed a temperature- and duration-dependent induction in the percentage and expression level of CAR in engineered T cells, with the highest induction observed at 43°C for 15 minutes (Figure S3E, S3F and S3G). CAR expression was detectable as early as 4 hours post-stimulation, with a peak observed at 6 hours. This rapid induction demonstrates the efficiency of the promoter in response to heat shock (Figure S3H). We then characterized and compared the *in vitro* killing capability of EchoBack-hGD2CAR (EchoBack CAR) and the previously published ultrasound-controllable CAR T cells (FUS-CAR) using co-culture assays.²³ The EchoBack-hGD2CAR T cells upon heat shock demonstrated significantly enhanced killing efficacy comparing to the original FUS-CAR T cells in all three donors that we have tested (Figure 3B). To mimic the *in vivo* tumor environment more accurately, we engineered a U-87 MG glioblastoma-derived three-dimensional (3D) tumor spheroid model.³⁸ This model enables the real-time monitoring of the interactions in the 3D tissue environment between tumor cells and EchoBack-CAR T cells expressing different fluorescent protein markers (Figure 3C). The EchoBack-hGD2CAR T cells, when activated twice by heat shock (HS 2X), almost entirely eliminated the tumor volume, in contrast to the minimal effect in the negative control without heat shock (Figure 3D and 3E, Video S1–3). While EchoBack-CAR exhibits a high sensitivity to heat shock, it has low leakage at rest or under temperatures of fever or hyperpyrexia (40°C), demonstrating its safety for the potential clinical applications (Figure S3E). Thus, EchoBack-CAR-T cells have a high killing efficacy against solid tumors while maintaining a safe profile.

EchoBack-hGD2CAR T cells eliminate the subcutaneous solid tumors *in vivo* without off-tumor toxicity

We then evaluated the *in vivo* tumor suppression of the FUS-controllable EchoBack-hGD2CAR T cells in subcutaneous glioblastoma model in mice. Specifically, engineered Fluc⁺ U-87 MG tumor cells were used to generate subcutaneous tumors, and EchoBack-hGD2CAR T cells were introduced into mice by local injection into the adjacent area of tumor and followed by a remote and noninvasive FUS activation (Figure 3F and S4A). The killing efficacy of these activated EchoBack-hGD2CAR T cells was evaluated by monitoring tumor growth and overall survival rate of the mice. To achieve better homing and residential time of CAR-T cells at the tumor site, we applied heat-shock to pre-induce the CAR expression in T cells *ex vivo* before injection into mice. These pre-induced EchoBack-CAR T cells (1X) affected the tumor growth moderately, while the injection of non-induced EchoBack-hGD2CAR T cells could not mitigate tumor growth (Figure 3G), which is consistent with our observations in the spheroid killing assay (Figure 3E). In contrast, these EchoBack-CAR T cells with additional two more FUS treatment significantly suppressed the tumor growth (Figure 3H), which led to the long-term survival of mice (Figure 3I).

EchoBack-hGD2CAR T cells further demonstrated stronger tumor killing *in vivo* than that of the FUS CAR²³ when both were exposed to FUS stimulation (Figure 4A–E). In fact, EchoBack-hGD2CAR showed a significant suppression of overall tumor growth across time (Figure 4B–D) and an elongation in survival time (Figure 4E). HE staining revealed no

detectable toxicity in major organs of mice treated by EchoBack-CAR, such as liver, lung, and kidney, verifying the safety of EchoBack-CAR (Figure S4C).

EchoBack-hGD2CAR outperforms the standard CAR in treating orthotopic glioblastoma

To examine the impact of the complex microenvironment of glioblastoma, we engineered orthotopic glioblastoma models and examined the antitumor efficacy of EchoBack-hGD2CAR T cells (Figure 4F). Orthotopic tumors were established by intracranial implantation of Fluc+ U-87 MG cells, followed by the intracranial administration of naïve T cells, standard hGD2CAR T cells, and EchoBack-hGD2CAR T cells,³⁹ which were subsequently exposed to FUS stimulations (Figure 4F). EchoBack-CAR T cells led to a substantial and enduring suppression of tumor growth, in contrast to the continuous tumor progression in the naïve T group and a transient antitumor response but a tumor relapse after 13 days in the standard hGD2CAR T group (Figure 4G and 4H). In addition, while mice treated with constitutive hGD2CAR T cells experienced a significant decline in body weight, indicating a systemic toxicity, the body weights of mice treated with either plain T cells or EchoBack-hGD2CAR T cells remained stable, with no significant reductions observed (Figure S4D). We performed a whole-brain assessment using H&E staining and found no damage or T cell infiltration in the normal brain (Figure S4E–P). These findings collectively confirm that EchoBack-CAR T cells provide effective tumor suppression with minimal off-tumor toxicity.

scRNAseq reveals the unique signaling network in EchoBack-CAR T cells

The anti-tumor differences between standard hGD2CAR and EchoBack-hGD2CAR T cells motivates us to characterize their functional properties. Exhaustion is one of the major challenges of CAR-T therapy in treating solid tumors and it has been shown that transient rest can restore the functionality of exhausted CAR-T cells.⁴⁰ Due to the inducible and reversible CAR expression in EchoBack CAR T cells, we hypothesize that the exhaustion profile of EchoBack CAR T cells could be different from that of the standard CAR-T cells where CAR is continuously expressed to engage tumor antigens. To test this hypothesis, we compared the transcriptomic profiles of EchoBack- and standard CAR T cells using single-cell RNA sequencing (scRNAseq), in which different T cells were repeatedly stimulated with fresh tumor cells daily for a period of 9 days to trigger chronic antigen activation on T cells (Figure 5A).

A total of 6,516 EchoBack-hGD2CAR and 13,017 standard hGD2CAR-T cells were obtained for transcriptomic profiling, achieving an average sequencing depth of 26,319 reads per cell. Unsupervised clustering analysis identified 8 unique clusters/subpopulations of CAR-T cells (Figure 5B). These groups exhibited distinct transcriptomic profiles, delineated by UMAP (Uniform Manifold Approximation and Projection) for visualization, with each cluster defined by its signature genes (Figure 5B and S5A–D). Notably, we found that clusters 0, 3, and 4, corresponding to memory CD8 T cells, cytotoxic^{high} effector CD8 T cells, and proliferating effector CD8 T cells, respectively, were markedly enriched in the EchoBack-CAR T group comparing to the standard CAR-T cell group (Figure 5B). This enrichment of CD8 T cells in EchoBack-CAR system was further confirmed in T cells derived from four additional donors (Figure S5E–H), underscoring the sustained cytotoxic

capability of EchoBack-CAR T cells following prolonged antigen exposure. Among the top significantly enriched feature genes within their respective clusters (Figure 5C), we observed a significant upregulation of Cyclic AMP-specific phosphodiesterases PDE4B and PDE3B,⁴¹ which are enzymes critical for degrading cAMP, a known inhibitor of acute T-cell activation in cluster 0. Cluster 3 was distinguished not only by the high expression of toxicity-related genes such as GNLY, GZMA, and NKG7 in its top ten genes, but also by a significant upregulation of ALOX5AP, which is a marker in CD8 lymphocytes with enhanced infiltration into the tumor microenvironment.⁴² Meanwhile, cluster 4 was characterized by an elevated expression of cytotoxicity genes alongside genes related to proliferation and the cell cycle, including MKI67, CENPF, and UBE2C, suggesting a dual role of these cells in mediating tumor destruction and supporting the proliferative expansion of the CAR-T cell population (Figure 5C). Comparing EchoBack-CAR and standard CAR T cells, pseudo-bulk analysis found that genes related to T cell survival and proliferation (e.g., PDE4B, DEC1, NFATC3), effector function (e.g., CD96, RUNX1), and infiltration (e.g., THEMIS, GRAP2) were significantly up-regulated, while genes associated with T cell inhibition/exhaustion (e.g., LGALS1, TIGIT) were downregulated in EchoBack-CAR T cells (Figure 5D). Through signaling pathway analysis, we found that functional interleukin signaling (e.g., IL-8, IL-7, IL-2, IL-6, and IL-1), mTOR signaling, JAK/STAT signaling, and GM-CSF signaling were collectively enriched in EchoBack-CAR T cells (Figure 5E). These results suggest a superior survival, proliferation, and effector functions of EchoBack-CAR T cells.

We also evaluated the cytotoxicity and exhaustion scores of different CAR T groups at the single-cell level. We found that the dominant clusters of EchoBack-CAR T cells showed significantly higher scores in cytotoxicity (Figure 5F and 5G) but lower in exhaustion (Figure 5H and 5I) than the standard CAR T group (Figure S5I), suggesting an upregulation in cytotoxicity alongside a down-tuning of exhaustion in EchoBack-CAR T cells. We also found that cytotoxicity-related genes (e.g., GZMA, GNLY, PRF1, IFNG, IL5) and exhaustion-related genes (e.g., LAG3, PDCD1, TIGIT, and CTLA4) were significantly upregulated and downregulated in EchoBack-CAR T cells, respectively (Figure S5J).

We further validated the scRNAseq results using ex vivo assays. At rest, standard CAR and FUS-inducible EchoBack-CAR T cells showed similar levels of exhaustion markers PD-1, Tim-3 and LAG-3 (Figure 5J, and S6). After 9 days of tumor stimulation, standard CAR T cells turned into exhaustion state more significantly as evidenced by a higher level of PD-1/TIM-3 or PD-1/LAG-3 double positive populations, comparing to the EchoBack-CAR T group (Figure 5K and S6). We also found that the difference in killing capability of pre-activated EchoBack-CAR T cells, and their standard counterparts was not significant before chronic antigen stimulation but became evident after chronic antigen challenges (Figure 5L). The above ex vivo exhaustion (Figure S6F and S6G) and cytotoxicity (Figure S6K) results were further confirmed in T cells derived from four additional donors. These results together demonstrate that EchoBack-CAR T cells exhibit high and sustained tumor-killing efficacy with reduced T cell exhaustion, possibly due to its nature of transient rest between FUS activations.

EchoBack-PSMACAR enables the robust elimination of prostate tumor

The modular design of EchoBack-CAR allows the targeting of different solid tumors. We explored this versatility of our EchoBack-CAR design by extending its application to the treatment of prostate-specific membrane antigen (PSMA) positive prostate cancer (EchoBack-PSMACAR). We also replaced the fluorescent marker with a clinically compatible and truncated form of CD19 (tCD19) to serve as an indicator of lentiviral transduction efficiency (Figure 6A). This tCD19 tag shortens the intracytoplasmic domain; as a result, tCD19 lacks active biological functions, and its expression does not interfere with the function of CAR and has been used as a marker in clinical trials⁴³. PSMA can be expressed in healthy tissues⁴⁴, such as proximal tubules in normal kidneys, the small intestine, and the bladder⁴⁵. Therefore, the use of standard PSMACAR T cell products could potentially lead to off-tumor effects on these healthy tissues, resulting in unforeseen side effects. We have tested the toxicity of standard CAR T cells that constitutively express PSMACAR, targeting primary human cells isolated from healthy tissues, and found that the standard CAR T cells can indeed cause significant toxicity against these healthy cells (Figure S6L), underscoring the critical need for additional safety controls in PSMA-targeted CAR-T therapies. We first validated that the EchoBack-CAR design could have a relatively sustained PSMACAR expression upon stimulation. At rest, EchoBack-PSMACAR T cells maintained a low level of CAR expression, even with the culture medium containing IL2, an activation cytokine of the feedback loop in EchoBack CAR T cells (Figure 6B). This finding is similar to what we have observed in EchoBack-hGD2CAR T cells (Figure S3E). In mice, we found ultrasound alone does not affect tumor growth compared to the plain T cell only group, indicating that ultrasound itself does not significantly contribute to tumor elimination (Figure S7A–C). After heat shock, the EchoBack-PSMACAR can induce a high CAR expression on the T cell surface, which gradually decays without tumor engagement. Remarkably, PSMACAR expression on surface can remain at a high level for up to five days when co-cultured with PSMA+ tumor cells (Figure 6B). The killing efficiency of the EchoBack-PSMACAR T cells was further tested *in vivo* (Figure 6C). Upon activation by focused ultrasound, the EchoBack-PSMACAR T cells exhibited potent and sustained cytotoxicity against PSMA-expressing prostate tumors, as demonstrated by a marked reduction in tumor volume (Figure 6D–E) and a prominent improvement in survival rates (Figure 6F) of the treated mice, comparing to the FUS-PSMACAR T cells or plain T cell group (Figure 6). The *in vivo* killing efficiency of EchoBack-PSMA CAR was also validated in T cells derived from an additional donor following the same treatment protocol (Figure S7D and S7E). These promising results not only underscore the efficacy of EchoBack-PSMACAR T cells in targeting and eliminating prostate cancer cells *in vivo* but also highlight the potential of EchoBack-CAR design as a modular and clinically relevant platform for the development of customized therapies for a broad range of solid tumors.

Reduced on-target off-tumor toxicity of EchoBack-CAR T cell therapy

While NSG mice are commonly employed in preclinical evaluations of CAR-T therapies due to their capacity to support human cell engraftment, their lack of fully functional human antigens limits their utility in assessing on-target off-tumor toxicities. To further demonstrate the safety and specificity of EchoBack-PSMACAR T cells, we employed a bilateral tumor model, with a distal tumor expressing low level of antigen mimicking the health tissue/organ

expressing similar antigens as the target tumor (Figure S7F–H). Our results showed that the standard PSMACAR T cells not only eliminated the proximal tumor, but also attacked and destroyed the distal tumor mimicking normal tissues expressing the same antigen after a delay of 12 days (Figure S7I and S7K). In contrast, EchoBack-PSMACAR T cells, after FUS activation, dramatically inhibited the growth of the proximal tumor while sparing the distal tumor (Figure S7J, S7L). These findings highlight the systemic activity and lack of specificity of standard CAR T cells, causing on-target but off-tumor toxicity. In contrast, the spatiotemporal control of the EchoBack-CAR T system led to localized anti-tumor activity while minimizing risks to normal tissues.

Discussion

We have developed an evolutionary sort-seq approach for promoter optimization to address the challenge of limited inducibility in heat shock promoters. The optimized promoter is further integrated with a genetic circuit of positive feedback loop that can respond to intrinsic CAR signaling and cause a more sustained CAR expression in engineered T cells, countering antigen-stimulated CAR endocytosis. Utilizing this approach, we engineered the EchoBack-CAR T cells for the treatment of solid tumors, which can sense the remote ultrasound signals and transduce into CAR production for therapeutic applications. Both the sonogenetic controllable EchoBack-hGD2CAR T and EchoBack-PSMACAR T cells developed in this study enable the robust suppression of glioblastoma and prostate solid tumors. In contrast, the standard hGD2CAR and PSMA CAR may pose toxicity risks to healthy tissues and hence human patients, as reported,^{33,45} thus precluding their direct use without additional safety controls. In high-fever mimicking situations, no significant killing of healthy human cells was observed with EchoBack-PSMACAR T cells, whereas standard constitutive PSMACAR T cells demonstrated significant cytotoxicity toward these same cell types (Figure S6L).

One of the significant challenges of standard CAR-T cell-based therapy for solid tumors, as demonstrated in multiple clinical trials,^{5–11} is the lethal on-target, off-tumor toxicity, mainly due to the candidate target antigens being expressed on non-malignant tissues.⁴⁶ This systematic challenge limits the dose escalation and other radical options/combinations needed in clinical trials for effective therapeutics. While the affinity of the scFv of the CAR with known off-tumor toxicity issues (e.g., ErbB2, EGFR, GD2) was reduced to spare normal cells expressing physiologic target,^{47,48} this approach can compromise the tumor-killing and therapeutic efficacy. Although low-affinity CAR-T cells were shown to mitigate tumors in rodent models, their effectiveness in human patients remains untested. Unlike the standard CARs with potential overactivation and off-tumor toxicities,^{20,49,50} the precise spatiotemporal control of EchoBack-CAR T cells with FUS provides a feasible solution to this problem. As the CAR expression will gradually decay when the EchoBack CAR T cells leave the local activation site at the tumor region, this will allow potent but less ideal CAR-antigen pairs to be used for effective tumor killing with minimal off-tumor toxicity. The EchoBack-CAR T cell further allows positive CAR feedback to enhance persistence and periodical rests to reduce exhaustion, verified by our scRNAseq results of the EchoBack-CAR T cells after chronic tumor stimulation.

The limited inducibility of the HSP promoter driving CAR is a significant hurdle in controllable CAR system.^{51,52} The application of high-throughput FACS combined with NGS (sort-seq) for the iterative counter-sorting of variants based on their performance under specific conditions (e.g., minimal leakage and high induction rates) is a strategy maximizing responsiveness to stimulant while minimizing undesired basal leakage of expression. As such, our sort-seq platform allows the optimization of the HSP promoter in a high throughput manner and is inherently modular for screening and optimizing other promoters. This platform can overcome drawbacks in traditional methods of promoter optimization focusing on predetermined sequences or rational designs, which may not fully capture the potential of inducible expressions.^{23,24,53}

In inducible systems, maintaining a more sustained CAR expression over prolonged periods is a critical challenge and can be addressed by the feedback design, as shown in the EchoBack-CAR design. By incorporating a self-sustaining feedback loop that harnesses intrinsic CAR activation signaling to promote further CAR production, this design can allow a sustained expression of CAR molecules in response to antigen presentation. Thus, this feedback loop offers a strategic solution to counter antigen presentation-induced CAR downregulation, improving the overall therapeutic efficacy of inducible CAR T cell therapies.^{31,54} This feedback mechanism can be integrated with the periodical FUS activations to mitigate T cell exhaustion by allowing intermittent “rest” periods, which are critical to the persistence and success of Echoback CAR-T cell therapy.⁴⁰ As we have shown in this study, the modular nature of this Echoback design extended its applicability across various CAR constructs and solid tumor targets, broadening the scope of CAR T cell therapy. In fact, the feedback network can be integrated with other costimulatory domains in CAR or independent armory molecules^{55–57} that activate NFAT, NF κ B, or cAMP signaling pathways, leveraging and extending the resonances between the CAR expression and activation signaling. With the evolving understanding of CAR signaling,⁴⁹ systematic characterization of natural or synthetic promoter sequences that can be activated by CAR and tumor antigen engagement is anticipated to continuously advance our EchoBack-CAR designs. As such, the modular platform integrating sort-seq for promoter optimization and feedback for CAR persistence under the remote and non-invasive FUS control can bridge engineering and medicine, leading to successful transitions from fundamental science and engineering to clinical applications.

Limitations of the Study

While this study primarily focuses on the molecular engineering of the EchoBack-CAR for ultrasound-controllable immunotherapy across multiple solid tumor models, we observed a preferential enrichment of CD8⁺ T cells in the EchoBack-CAR system compared to standard CAR-T cells (Figure S5), potentially due to reduced T cell exhaustion in EchoBack-CAR T cells. The further investigation is needed to elucidate the molecular mechanisms underlying this CD8⁺ preferential enrichment. Locoregional delivery of CAR-T cells has been increasingly applied in clinical settings.^{58–60} Similarly, the EchoBack-CAR T cells demonstrated the best efficacy with localized delivery at the current stage and are expected to require higher cell numbers for systemic administration. Future engineering of EchoBack-CAR T cells for systemic delivery and enhanced therapeutic efficacy may

involve incorporating tumor-specific homing receptors or tumor-targeting motifs.^{61–63} NSG mice, commonly used for preclinical evaluations of CAR-T therapies due to their ability to support human cell engraftment, lack functional human antigens and an intact immune system, limiting their utility for assessing potential CAR-T cell toxicities. The evaluation and characterization of EchoBack-CAR T cells in more biomimetic models or clinical trials will be essential to comprehensively evaluate their safety and behavior under complex physiological conditions.

Resource availability

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Yingxiao Wang (ywang283@usc.edu).

Materials availability

All resources, materials and reagents generated in this study are available from the lead contact with a completed Materials Transfer Agreement.

Data and code availability

Raw and processed sequencing data for this study can be accessed in the NCBI Gene Expression Omnibus (GEO) database under the accession number GSE289368. The code repository for the PID controller of the in-house built FUS system has been published in our previous study⁶⁴.

STAR Methods

Experimental model and study participant details

Tissue culture: Human Glioblastoma cell U-87 MG Malignant Glioma (U-87 MG) and human embryonic kidney HEK293T cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum (FBS) and 1% penicillin–streptomycin (P/S). PC-3 and Jurkat cells were cultured in Roswell Park Memorial Institute Medium (RPMI 1640) with 10% FBS and 1% P/S. Human small intestine epithelial cells (HIEC-6) were cultured in OptiMEM Reduced Serum Medium supplemented with 10 ng/mL Epidermal Growth Factor (EGF) and 4% fetal bovine serum (FBS). Human Primary Bladder Epithelial Cells (BdEC) were cultured in Bladder Epithelial Cell Basal Medium supplemented with Bladder Epithelial Growth Kit. Primary human T cells were cultured in X-VIVO 15 supplemented with 5% FBS, 1% P/S, 50μM 2-Mercaptoethanol, and approximately 100 U/ml recombinant human IL-2. Cells were cultured at 37°C in a humidified 5% CO2 incubator unless Otherwise Mentioned.

Animal models: Male NSG mice (NOD.Cg-Prkdc^{scid} Il2rgt^{m1Wjl}/SzJ) mice of 6–8 weeks old were purchased from Jackson Laboratory. Animal studies were approved in Protocol S15285 and 21479 approved by the UCSD and USC Institutional Animal Care and Use Committee (IACUC), respectively. All researchers complied with animal-use guidelines

and ethical regulations during animal studies. Mice were housed in sterile cages in barrier facilities at UCSD and USC.

Source of primary human T cells: Human peripheral blood mononuclear cells (PBMCs) were isolated from buffy coats obtained from the San Diego Blood Bank using lymphocyte separation medium. Primary human T cells were subsequently isolated from PBMCs using the Pan T Cell Isolation Kit. Isolated T cells were cryopreserved in 7.5% DMSO + 92.5% FBS until use.

Method details

HSE library: The HSE promoter library plasmid was constructed through a two-step cloning process. Initially, DNA comprising an Enhanced GFP (EGFP) and a 1HSE-YBTATA promoter was amplified via polymerase chain reaction (PCR) and cloned into the pHR lentiviral vector to form pHR-1HSE-EGFP-PGK-mCherry construct. Subsequently, the HSE-library-YBTATA-EGFP fragment was amplified by PCR, during which mutagenesis was introduced through NNG primers bearing random mutations at designated HSE positions. ESP3I cutting sites were introduced at both ends of the fragment through the primers. The HSE-library-YBTATA-EGFP fragment was further cloned into the lentiviral vector pLVX by replacing the LacZ gene in pLVX-(ESP3I) LacZ (ESP3I)-SFFV-mCherry via Golden Gate assembly. Cloning was performed using Gibson assembly. Primers utilized in this study were synthesized by Integrated DNA Technologies (IDT, Coralville, IA). Seven repeats of HSEs were synthesized by Genewiz (Azenta Life Sciences, South Plainfield, NJ). PCR reactions were conducted using Q5 DNA Polymerase, and sequences were verified by Sanger sequencing (Azenta Life Sciences, South Plainfield, NJ).

Heat shock of cells in vitro: Jurkat or human primary T cells were centrifuged and resuspended in the corresponding medium at a concentration of 10–20 million cells per ml (0.5–1M/50µl) and then aliquoted into PCR 8-strip tubes. The cells were placed into a PCR thermocycler to undergo a 2-minute prewarm at 37°C, followed by heating at the desired temperature/time. After heating, cells were diluted back to a normal density (0.5–1M/ml) and returned to standard culture conditions.

Lentiviral production: The plasmid of interest, 2nd generation lentiviral packaging plasmid pMD2.g, and PsPax2 were co-transfected into Lenti-X 293T cells using either the ProFection Mammalian Transfection System or Lipofectamine 3000 Transfection Reagent following the manufacturer's protocol. 48 hours after transfection, the supernatant was collected and filtered through 0.45 µm PES or PVDF filters. If necessary, the virus was further concentrated using Lenti-X concentrator following the manufacturer's protocol. A specific amount of virus was then used to infect Jurkat cells. Functional titers (transduction units) were calculated based on the infection positive rate measured by flow cytometry three days after adding the lentivirus.

T cell library and screening via FACS: Library plasmids were utilized to generate a lentiviral library and were subsequently transduced into Jurkat cells at a multiplicity of infection (MOI) of 0.05. Cells infected with library constructs were enriched via

fluorescence-activated cell sorting (FACS) using the mCherry marker until enough cells were obtained to achieve 10X coverage of the library. In this scenario, it is expected that >97% of the sorted cells would contain only one variant of the library. Jurkat cells containing the HSE library variant were subjected to heat shock at 43°C for 15 minutes. 6 hours after heat shock, cells exhibiting a high EGFP/mCherry intensity ratio (top 20%) were enriched via FACS. The enriched cells were then cultured under normal conditions for 5–7 days to allow EGFP levels to recover to basal levels. Subsequently, another round of FACS was performed to sort out cells with negative EGFP expression, thereby enriching for cells with low basal leakage. The cycles of heat shock induction followed by sorting for low leakage were repeated as necessary.

Illumina DNA sequencing of biosensor variants: Following each round of FACS, genomic DNA from collected cells was extracted using the Zymo Quick-DNA Microprep Kit. Sequences within the HSE region were amplified for Illumina sequencing using a low cycle PCR (25 cycles) with adaptor-containing primers customized for each round of screening (IDT, IDT, Coralville, IA; Table S1). The sizes of the amplicon libraries were confirmed via gel electrophoresis, and the amplicons were subsequently purified using the Zymoclean Gel DNA Recovery Kit. Samples were then sequenced using the Illumina HiSeq 4000 sequencing system with PE 150 sequencing at the IGM Genomic Center, UCSD.

The sequencing data underwent analysis using code written in R. Sequences with a Phred score >20 across all positions covering the constant regions were retained. To address potential biases arising from sequencing depths, the frequency of unique sequences (fv) was calculated by normalizing the variant count in each library group to the total number of sequences reads for that group. This normalization was achieved by converting the read counts of each sequence to Counts Per Million (CPM), accurately reflecting the frequency of a unique sequence. Sequences with CPM values exceeding 10 were considered positive and selected for further analysis. Importantly, since different libraries had varying total sequencing reads, the frequency of a unique sequence (fv) allowed for comparisons across different library groups. The fv from the final rounds of screening was used to rank the candidates.

Human Primary T cell transduction: The isolated primary human T cells were activated with Dynabeads Human T-Expander CD3/CD28 at a 1:1 or 2:1 cells-to-beads ratio. T cell spinoculations were conducted 48 hours after Dynabeads activation, during which activated T cells were mixed with lentivirus at a multiplicity of infection (MOI) of 5–10 and then placed in well plates precoated with Retronectin, followed by centrifugation at 1800g at 32°C for 1 hour. Dynabeads were removed 5 days after initial activation, followed by FACS if necessary. T cells were further expanded until day 7–15 after activation and either frozen or used for experiments, unless otherwise specified.

Cell staining and flow cytometry. Multiple fluorophore-conjugated antibodies were employed in this research for surface protein staining. Cells were washed and resuspended in 100 µl staining solution (PBS containing 1% FBS) with the recommended amounts of antibodies and incubated for the suggested duration (according to the manufacturers' instructions) on ice. Following incubation, samples were washed three times before flow

cytometry analysis/FACS (BD Accuri C6 or SONY SH800). The FCS data were analyzed using FlowJo software.

Luciferase-based cytotoxicity assay: Unless otherwise specified, 1×10^4 Fluc+ U-87 MG cells were seeded into 96-well plate 3 hours prior to adding T cells. Engineered primary human T cells with or without heat shock (HS) were added to the tumor cells at different effector-to-target (E:T) ratios or no T cells 2 hours post HS. The mixtures were then cultured for the desired duration (24 h, 48 h) and assayed using the ONE-Glo Luciferase Assay System following the manufacturer's instructions to quantify luminescence. For HIEC-6 and BdEC cytotoxicity assays, 1×10^4 Fluc+ HIEC-6 or BdEC were seeded into 96-well plate 3 hours prior to adding T cells. Engineered standard or EchoBack PSMA CAR-T cells were then added to the target cells at an effector-to-target (E:T) ratio of 1:1. For the EchoBack groups, the co-cultured cells were incubated at 41°C for 4 hours, followed by culturing at 37°C until 48 hours post T cell addition and assayed using the ONE-Glo Luciferase Assay.

3D tumor spheroids assay: U-87 MG cells were utilized in the 3D tumor spheroid printing process following the published literature³⁸. Briefly, a magnetic device was assembled with corresponding μ -Plate 96 Well Square plates to assist the uniform tumor spheroids formation. Subsequently, a cell suspension in DMEM with 10% FBS and 1% P/S, supplemented with 25mM Diethylenetriaminepentaacetic acid gadolinium (III) dihydrogen salt hydrate (97%), was added to each well. The entire system was then incubated at 37°C for 3 hours to facilitate spheroid formation. The system is then transferred into Incucyte S3 for further experiments. If needed, 3D tumor spheroids were heated in water bath at desired temperature/duration.

Repeated antigen stimulation assay: Standard or EchoBack-hGD2 CAR T cells were co-cultured with U-87 MG glioblastoma cells at a 1:2 effector-to-target (E:T) ratio in a mixture of 50% X-VIVO (5% FBS) and 50% DMEM (10% FBS), supplemented with approximately 100 U/ml recombinant human IL-2. Every 48 hours, supernatant from the co-culture plate containing CAR-T cells was collected and centrifuged. The cell pellet was then subjected to heating at 43°C for 15 minutes, and live cells were counted using AOPI live-dead staining. Subsequently, 0.125 million CAR-T cells were seeded. Fresh U-87 MG cells (0.25 million) were added to the T cells daily to maintain repeated stimulation. Cells were collected on day 9 of the assay and sorted for T cells. Purified T cells were used for cytotoxicity and exhaustion marker assays or frozen for further single-cell RNA sequencing.

Single cell-RNA sequencing: CAR-T cells from the chronic antigen stimulation assay were processed for subsequent single-cell RNA sequencing using the Chromium platform (10X Genomics) and the Chromium Single Cell 3' HT v3.1 kit at the Molecular Genomics Core, USC Norris Comprehensive Cancer Center. cDNA libraries were separately prepared for EchoBack CAR-T cells and standard CAR-T cells. Sequencing was performed on the Illumina NextSeq 2000 system. A total of 6,516 HS-CAR and 13,017 PGK-CAR T cells were identified, yielding an average of 26,319 reads per cell. Alignments and count aggregation of gene expression and sgRNA reads were completed using Cell Ranger (v.7.0.1). Gene expression and sgRNA reads were aligned using the Cell Ranger count

command with default settings. Gene expression reads were aligned to the human (hg38) reference genome using STAR 2.7.2a. A sample-specific approach was employed to identify the boundary between empty microwells and those containing a cell, based on log-scale barcode rank plots. Low-quality cells with <200 detected genes or >15% mitochondrial reads were discarded. Cells with fewer mRNA UMIs than the specified threshold were excluded from downstream analysis. Quality control criteria for unsupervised clustering involved retaining cells with gene expression counts between 1,000 and 6,500 genes. Subsequently, mRNA expression data were normalized using the 'SCTransform' function in Seurat,⁶⁵ and batch effects related to experimental conditions were removed via the Seurat v3 anchor-based integration workflow using default parameters. Finally, the integrated mRNA data were visualized in two dimensions using UMAP. Differentially expressed genes (DEGs) were identified using the 'FindMarkers' function in Seurat with a log fold-change threshold of 0.25. Ingenuity Pathway Analysis (IPA) was employed to reveal signaling pathways linked to each cluster. The top 12 T cell-related pathways were selected based on the mean activation z-score ($z < 0$ indicating inhibition, $z > 0$ indicating activation, and $|z| \geq 1.645$ considered significant).

FUS system: The Focused Ultrasound (FUS) system equipped with a real-time temperature control feedback loop was utilized for conducting hyperthermia experiments. The system comprises a custom-made 1.1-MHz single-element transducer, fabricated in-house using a pre-focused modified PZT material, with specific dimensions (diameter: 70mm, radius of curvature: 65mm) and a 20mm hole in the center. To facilitate the transmission of ultrasound waves and guide the focus, a 3D-printed coupling cone (length: 65mm) with a 4mm diameter opening at the tip was utilized and attached to the transducer. This cone was designed to hold degassed water along the acoustic path, with the opening at the tip sealed with an acoustically transparent thin film. Deionized water was degassed using a vacuum pump to remove air bubbles and ensure optimal acoustic transmission. The transducer was driven by a function generator and a 50dB power amplifier, delivering pulsed sine waves to the transducer.

In vivo tumor models and FUS stimulation: For subcutaneous tumor models, 1×10^6 Fluc+ U-87 MG or 2×10^5 Fluc+ PSMA+ PC-3 cells were injected subcutaneously into NSG mice on Day 0. Tumor growth was monitored by in vivo bioluminescence imaging. 1×10^6 Engineered primary human T cells were administered intratumorally on day 5–6. For bilateral subcutaneous tumor models, NSG mice were injected with 0.2×10^6 PSMA^{High} PC-3 cells into the proximal flank and 0.2×10^6 PSMA^{Low} PC-3 cells into the distal flank. 1×10^6 Engineered primary human T cells were locally injected only into the proximal tumor on day 7. FUS stimulations targeting the tumor region were performed at planned time points. For intracranial tumor models, 2×10^5 Fluc+ U-87 MG cells were injected intracranially. The mouse was anesthetized with isoflurane and placed onto a stereotaxic instrument. Following skin preparation and disinfection, a scalp incision was made starting behind the right eye and extending to a length of 10 to 15 mm. The injection site was marked as a point 1 mm above and 2 mm lateral to the bregma. A syringe equipped with a 32-gauge needle was used to load 5 microliters of suspended Fluc+ U-87 MG cells, which were then injected to a depth of 3 mm at the target site. Following injection, the skin was sutured

closed, and the wound site was sterilized with iodine. 2×10^6 Engineered primary human T cells were administered intratumorally on day 6–7 at the tumor site. FUS stimulations targeting the tumor region were performed at planned time points. For FUS stimulation targeting subcutaneous tumor models, mice were anesthetized and placed on a heating pad set to 37°C to maintain body temperature. To control the heating temperature accurately, a needle-type thermocouple that connected to a thermometer was inserted into the tumor site. To conduct FUS stimulation targeting intracranial tumor models, mice were anesthetized and positioned on a stereotaxic instrument. The FUS stimulation target remained consistent with the injection target. The FUS transducer was positioned to focus on the tumor, with acoustic gel applied between the mouse skin and the transducer. A PID controller, receiving real-time temperature feedback, adjusted the output power of the function generator to maintain the focal temperature at approximately 43 °C.

Mouse Tissue H&E Staining and Immunohistochemistry Staining: Mouse brains and major organs were collected and fixed in 10% formalin at the endpoints of each experiment. Tissues were embedded in paraffin and sectioned for H&E staining by the Liver Histology Core of the USC Research Center for Liver Diseases or by the USC Mann Histology Laboratory. For IHC staining, paraffin-embedded brain tissue sections were deparaffinized, rehydrated, and subjected to heat-induced epitope retrieval using citrate buffer. After blocking with BSA, the sections were incubated overnight at 4°C with an anti-CD8 primary antibody. The next day, an HRP-conjugated secondary antibody was applied for 1 hour at room temperature. Signals were visualized using the Pierce DAB Substrate Kit, followed by counterstaining with hematoxylin, dehydration, and mounting for subsequent analysis.

In vivo bioluminescence imaging: Firefly luciferase substrate D-luciferin was administered intraperitoneally, followed by In vivo bioluminescence imaging (BLI) approximately 8–12 min later. Signal is captured by Lumina LT Series III. Images with peak signals were analyzed with Living Image software. Tumor size is analyzed based on the integrated reading within a fixed region of interest (ROI) over the tumor.

Quantification and Statistical Analysis: Quantification results were analyzed in GraphPad Prism (GraphPad Software) to test statistical significance with the respective tests. Detailed statistical information for each figure is provided in the corresponding figure captions.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Research Highlights:

1. Sonogenetic EchoBack-CAR T cells eliminate solid tumors with minimal toxicity.
2. An ultra-sensitive FUS-inducible promoter has been identified from library screening.
3. A feedback loop reprograms tumor engagement into CAR production upon FUS stimulation.
4. EchoBack-CAR T cells show stronger cytotoxicity and less exhaustion than standard CARs.

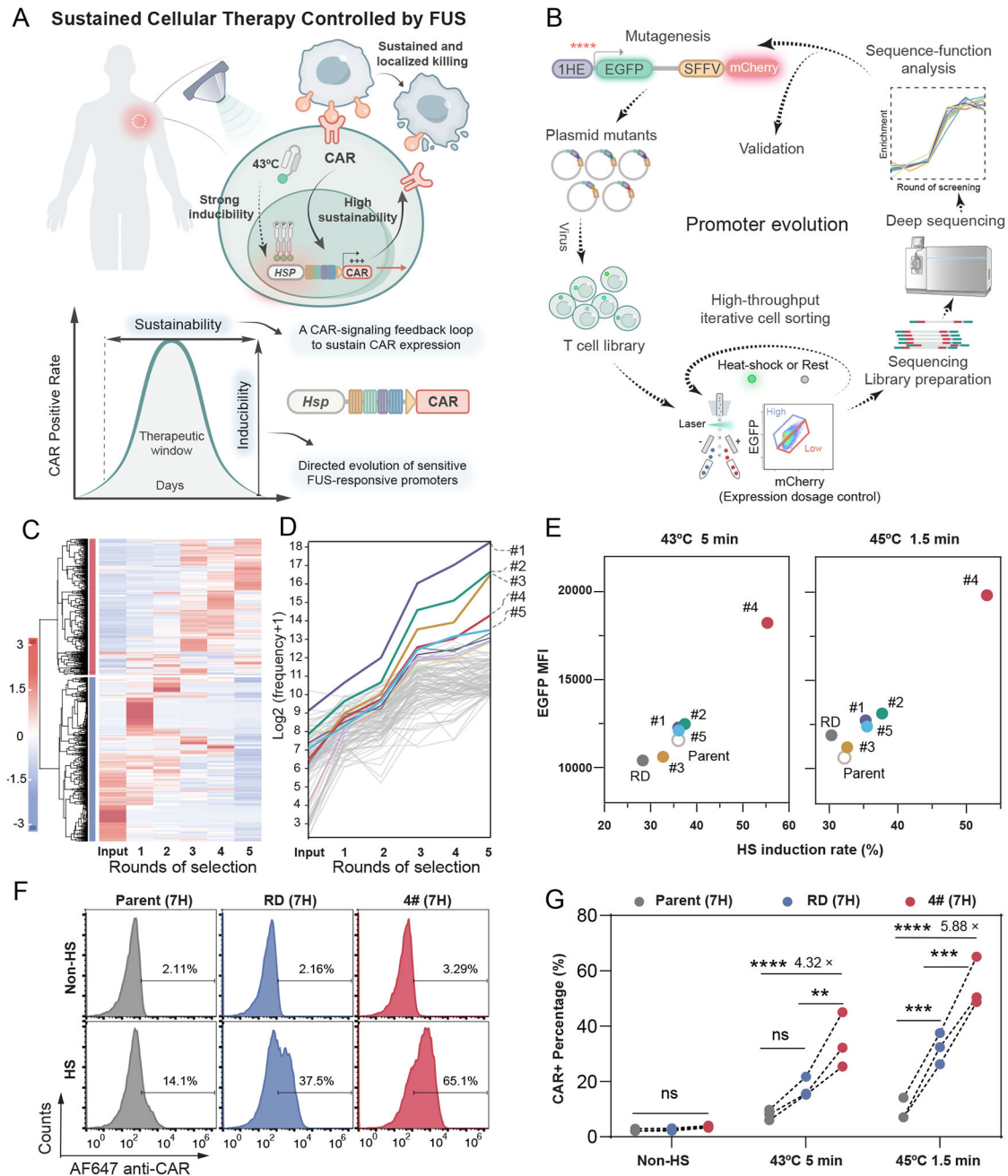


Figure 1. Identify potent heat-sensitive promoters via sort-seq.

A. Graphical overview of the focused ultrasound (FUS)-controlled sustained cellular therapy for solid tumor. In this therapy, FUS serves as an ‘on switch’, initiating CAR expression for solid tumor therapy. The concept of a “therapeutic window” is defined by the period after a single FUS application where CAR expression exceeds a pivotal threshold, facilitating efficient tumor cell eradication. To increase inducibility, we employed an evolutionary sort-seq approach, integrating interactive cell screening and sorting with next-generation sequencing. This strategy aimed to identify a highly sensitive heat-shock responsive element optimized for FUS-induced thermal activation. By further leveraging the activation of the

intrinsic CAR signaling pathway and incorporating its corresponding responsive elements, a positive feedback loop was engineered that converts CAR activation signaling into sustained and enhanced CAR production.

B. A scheme of the evolutionary sort-seq approach to identify potent heat-sensitive promoters. The library DNA, encompassing all mutant promoter possibilities, was transduced into Jurkat T cells to create a mammalian T cell library. Multiple rounds of counter-sorting enrichment were performed to isolate cells hosting high induction and low leakage variants, with the sequences identified using next-generation sequencing (NGS). 1HE: single heat shock element; SFFV: spleen focus-forming virus promoter.

C. Heatmap depicting the progressive enrichment and depletion of mutant sequences during screening. Each row represents one mutant sequence, color-coded by its enrichment score, which is normalized as a z-score for illustration purposes. Clustering was determined using k-means.

D. The enrichment frequency change of the top 100 HSP mutant candidates is depicted after each sorting round, with the top 10 candidates color-coded. The top 5 candidates were selected for further validation.

E. Heat-responsive potency of selected candidates. The 1HE-EGFP constructs were tested in Jurkat T cells 6 hours post-heat shock at 43°C for 5 minutes (left) and 45°C for 1.5 minutes (right). EGFP mean fluorescent intensity (MFI) (Y-axis) and the percentage (X-axis) of the top 5 candidates were measured and compared to the parental control. RD, rationally designed 1HE; parent, Hsp 70.1 1HE. The results are representatives of 4 independent experiments.

F. Flow cytometry histograms displaying CAR expressions with or without induction upon a 45°C, 1.5 minutes heat shock (HS). Parent (7H), RD (7H), and 4# (7H) refer to seven repeats of parental, rationally designed, #4 HSE of figure E.

G. CAR expression in Jurkat T cells measured 6 hours following various heat shock conditions. From left to right: NS, not significant $p=0.9983$, ns $p=0.0933$, **** $P<0.0001$, ** $P=0.0028$, *** $P=0.0001$, *** $P=0.0001$ and **** $P<0.0001$ respectively. $n=3$ in each group.

See also Figure S1.

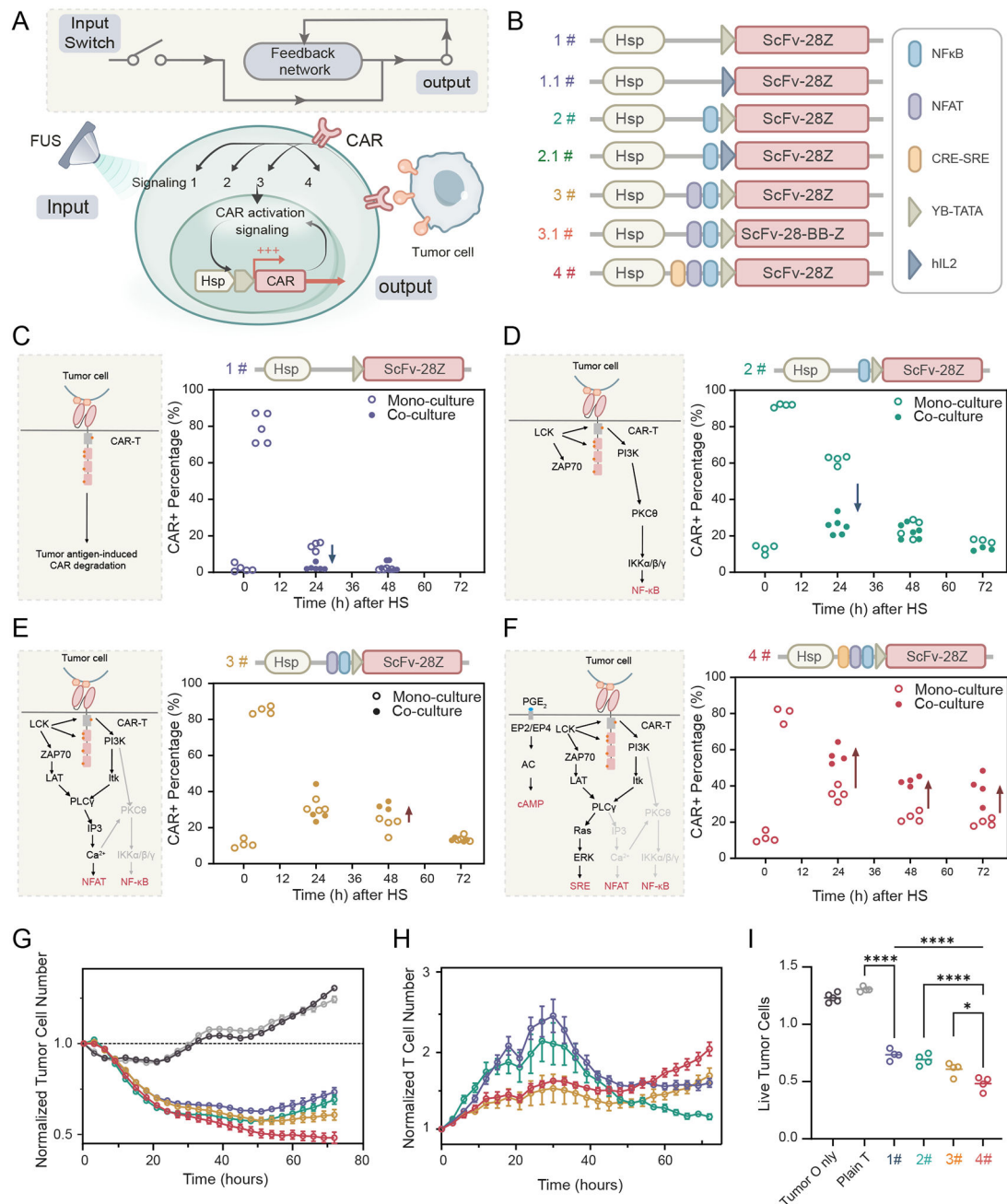


Figure 2. Design a positive feedback loop to enhance the sustainability of CAR functions.

A. Schematic illustration of the feedback loop design for sustained CAR expression. Focused ultrasound (FUS) serves as an 'on switch', inducing the CAR expression, which will be activated upon antigen engagement to trigger a positive feedback loop to convert CAR signaling into sustained CAR production.

B. The different feedback CAR designs. HSP, synthetic heat shock promoter with 7 repeats of HSEs; NFκB: NFκB responsive element; NFAT: NFAT responsive elements; CRE: cAMP responsive elements; SRE: serum responsive elements; YB-TATA: YB-TATA minimal promoter; hIL2: human IL-2 minimal promoter.

C-F, Kinetics of hGD2CAR expression before or after heat shock activation in Jurkat T cells with various positive feedback loop designs, assessed in the co-culture of tumor cells expressing the antigen (solid dot) and in monoculture without tumor cells (open dot). A simplified CAR signaling pathway diagram in T cells is presented on the left. Sonogenetic CAR-T cells were subjected to heating at 43°C for 15 min. CAR expression levels were initially assessed 6 hours after heat stimulation to determine inducibility, followed by co-culture with U-87 MG glioblastoma cells at an E: T ratio of 1:1. The CAR molecules were identified using the AF647-labeled anti-CAR antibody. The blue and red arrows represent CAR reduction and increase upon antigen engagement, respectively. At least n=3 biological replicates were included (n is shown in the figure).

G-H. Dynamics of glioblastoma U-87 MG cells (G) and CAR-T cells (H) in the IncuCyte live-cell assay. CAR-T cells were subjected to heating at 43°C for 15 minutes before co-culture. U-87 MG or CAR-T cells were labeled with EGFP or mCherry, respectively. N=4 for each group. Error bars, Mean $\pm$ SEM.

I. Quantification of relative live tumor cell numbers at 72 hours post-co-culture with CAR-T cells incorporating various feedback loop designs. n=5 in the tumor-only group and n=4 in all other groups. one-way ANOVA, ****p<0.0001 and *p=0.015, respectively. See also Figure S2.

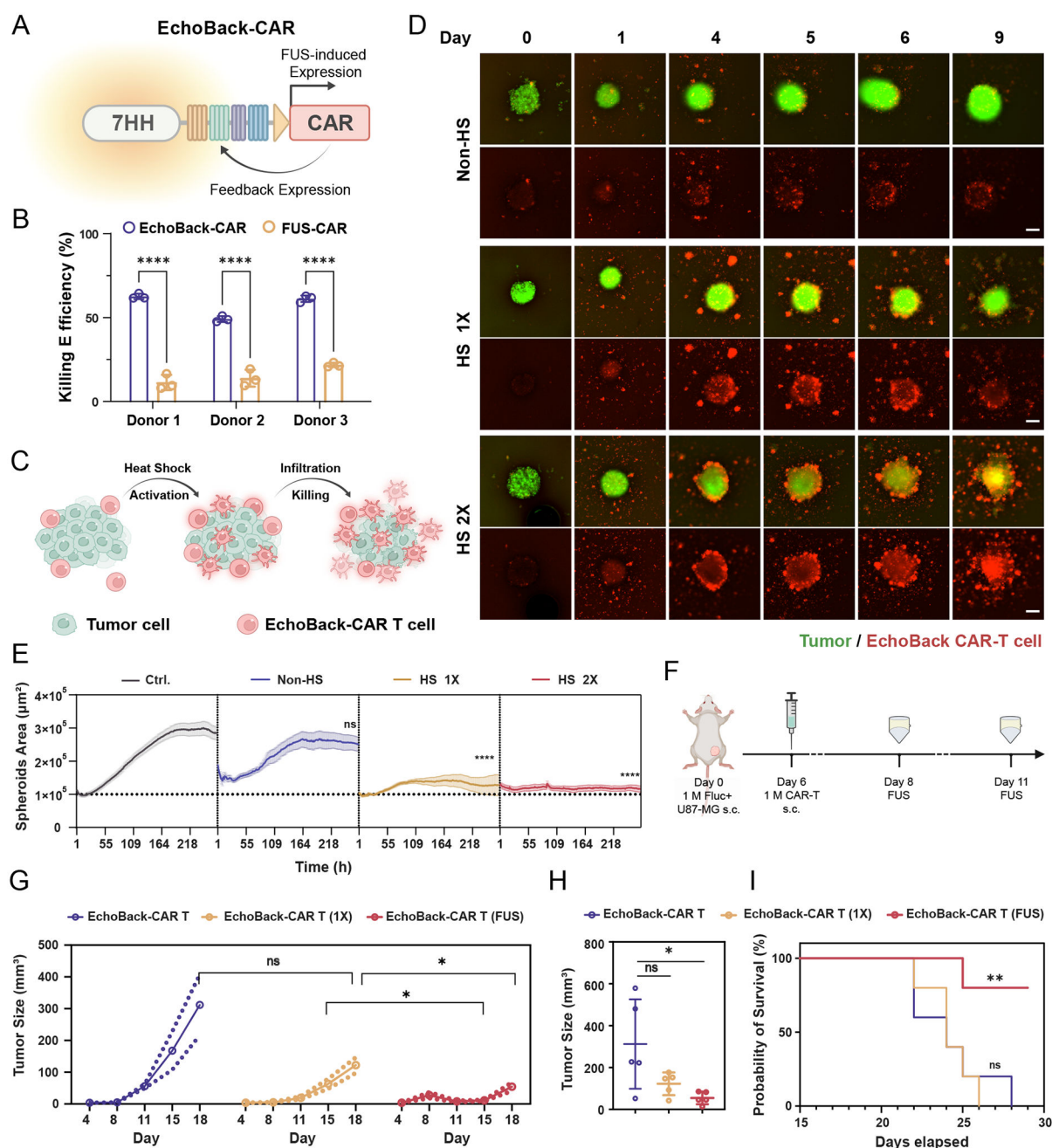


Figure 3. Strong killing efficacy of EchoBack-CAR T cells against tumor cells in vitro and in vivo.

A. Schematic illustration of EchoBack-CAR. In EchoBack-CAR T cells, CAR expression is driven by a hybrid promoter that combines a high-induction heat shock promoter, optimized through evolutionary sort-seq, with elements from the CAR signaling feedback loop.

B. Killing efficiency of EchoBack-CAR-T cells versus control FUS-CAR T cells in co-culture with U-87 MG glioblastoma cells. The CAR-T cells were treated with heat shock at 43°C for 15 minutes and then co-cultured with Fluc⁺ GD2⁺ U-87 MG cancer cells at an E:T ratio of 1:4. Two-way ANOVA. Error bars, mean $\pm$ SD. ****p < 0.0001, n=3 in each group.

C. Schematic illustration of the activation mechanism of EchoBack-CAR T cells and their efficacy in targeting solid tumors or 3D tumor spheroids. Upon heat-shock stimulation, these CAR-T cells express CAR and become activated upon engagement with tumor cells. This interaction triggers a prolonged CAR expression through CAR signaling feedback, facilitating enhanced T-cell infiltration into and clearance of tumor spheroids.

D. Representative fluorescent images showing tumor spheroids (green, EGFP+) and EchoBack-CAR T cells (red, mCherry+) under various heat shock stimulation conditions. Scale bar=200 μ m.

E. Quantification of U-87 MG glioblastoma spheroid area in the absence (Control, Ctrl.) or presence of co-cultured EchoBack-hGD2CAR T cells under various heat shock stimulation conditions. Non-HS, non-heat shock; 1X HS, 43°C 15 minutes heat shock was applied on day 2; 2X HS, 43°C 15 minutes heat shock was applied on day 2 and 5. One-way ANOVA. Control group was used as reference. From left to right, ns $p = 0.7877$, **** $p < 0.0001$ and **** $p < 0.0001$ and $n = 6, 8, 9, 9$, respectively. Error bars, mean $\pm$ SEM.

F. Experimental timeline for EchoBack-hGD2CAR T therapy in NSG mice. Fluc+ U-87 MG cells were subcutaneously engrafted into NSG mice, followed by the administration of EchoBack-CAR T cells and localized FUS stimulation at the tumor site to assess in vivo efficacy. FUS stimulations were performed at 43°C for 10 minutes directly at the tumor site.

G. Tumor growth was monitored using caliper measurements in each experimental group. Two-way ANOVA. From left to right, ns $p = 0.0503$, * $p = 0.0264$, and * $p = 0.0489$. $n = 5$ mice per group. Error bars, mean $\pm$ SEM.

H. Tumor size on day 18 as in (B). Error bars, mean $\pm$ SEM. One-way ANOVA. NS, Not Significant $p = 0.0885$; * $P = 0.0204$.

I. Survival curve of mice treated with EchoBack-CAR T cells across varied stimulation conditions. Log-rank (Mantel-Cox) test; ** $p = 0.0086$; ns, not significant $p = 0.8318$. See also Figure S3 and Figure S4.

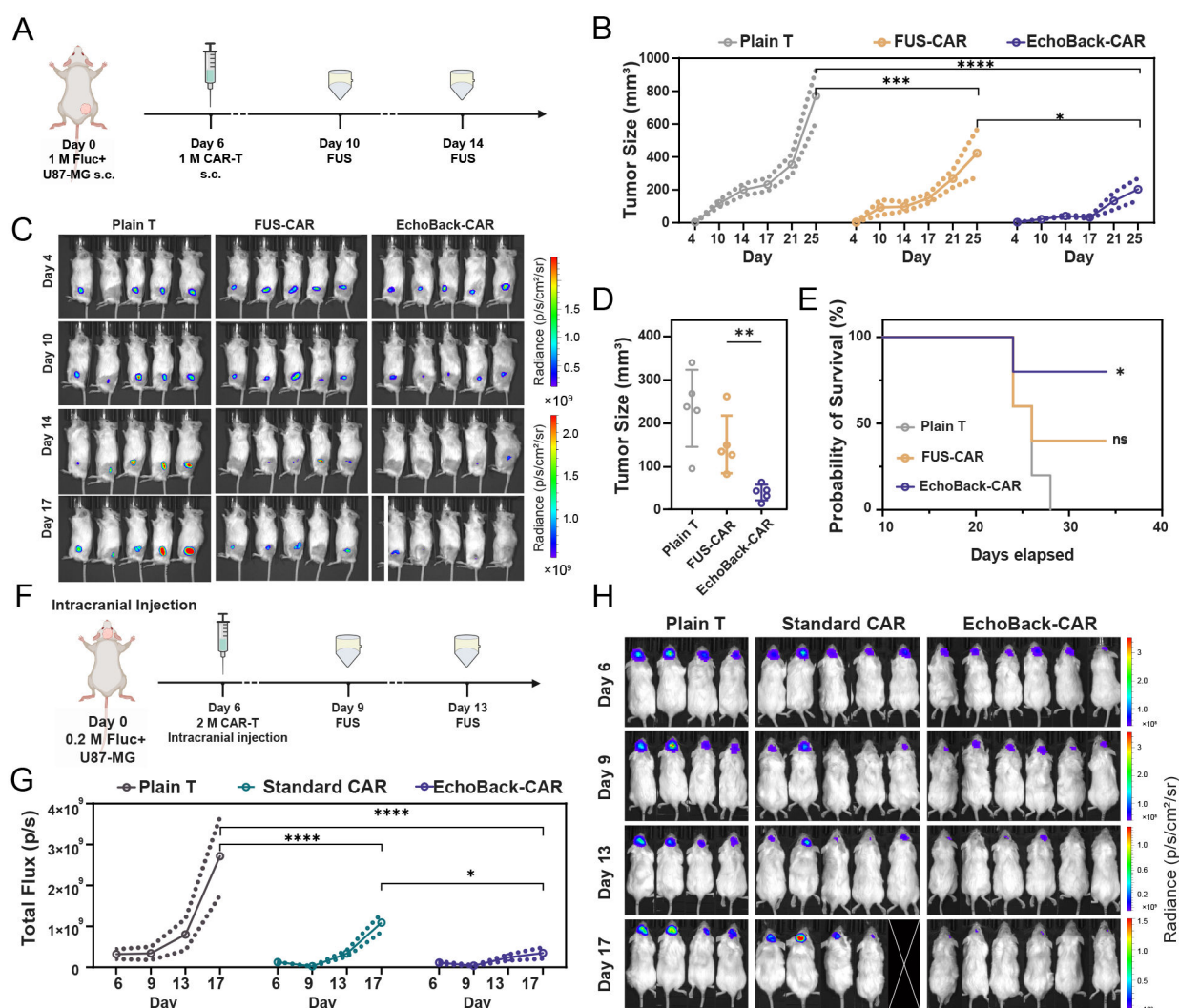


Figure 4. EchoBack-hGD2CAR T cells eliminate the subcutaneous and orthotopic glioblastoma in vivo.

A. Experimental timeline of EchoBack-hGD2CAR T and FUS-hGD2CAR therapy in NSG mice. The mice in the FUSCAR and EchoBack-CAR group received the FUS stimulation.

B. Tumor growth was monitored using caliper measurement. Two-way ANOVA. From left to right, ***p= 0.0002, ****p<0.0001, and *p=0.0226. n=5 mice in each group. Error bars, mean ± SEM.

C. Bioluminescence images of the mice in (B).

D. Tumor size on day 17 as in (B). Error bars, mean ± SEM. One-way ANOVA. **P = 0.007.

E. Survival curve of mice treated with EchoBack-hGD2CAR T cells or FUS-hGD2CAR T cells. n=5 mice in each group. Log-rank (Mantel-Cox) test; NS, not significant p= 0.3133; *P = 0.0228.

F. Timeline of the orthotopic glioblastoma treatment experiment: the orthotopic model was established by intracranial injection of Fluc+ U-87 MG cells, followed by the intracranial

administration of either standard hGD2CAR or EchoBack-hGD2CAR T cells for treatment. The mice in the EchoBack-hGD2CAR group received the FUS stimulation.

G-H. Tumor growth in the orthotopic glioblastoma model measured by bioluminescence imaging(G), and its representative images (H) for mice in different treatment groups.

Two-way ANOVA. From left to right, **** $p < 0.0001$, **** $p < 0.0001$, and * $p = 0.0158$, respectively. From left to right, $n = 5, 10, 8$ respectively in H. Error bars, mean $\pm$ SEM.

See also Figure S4.

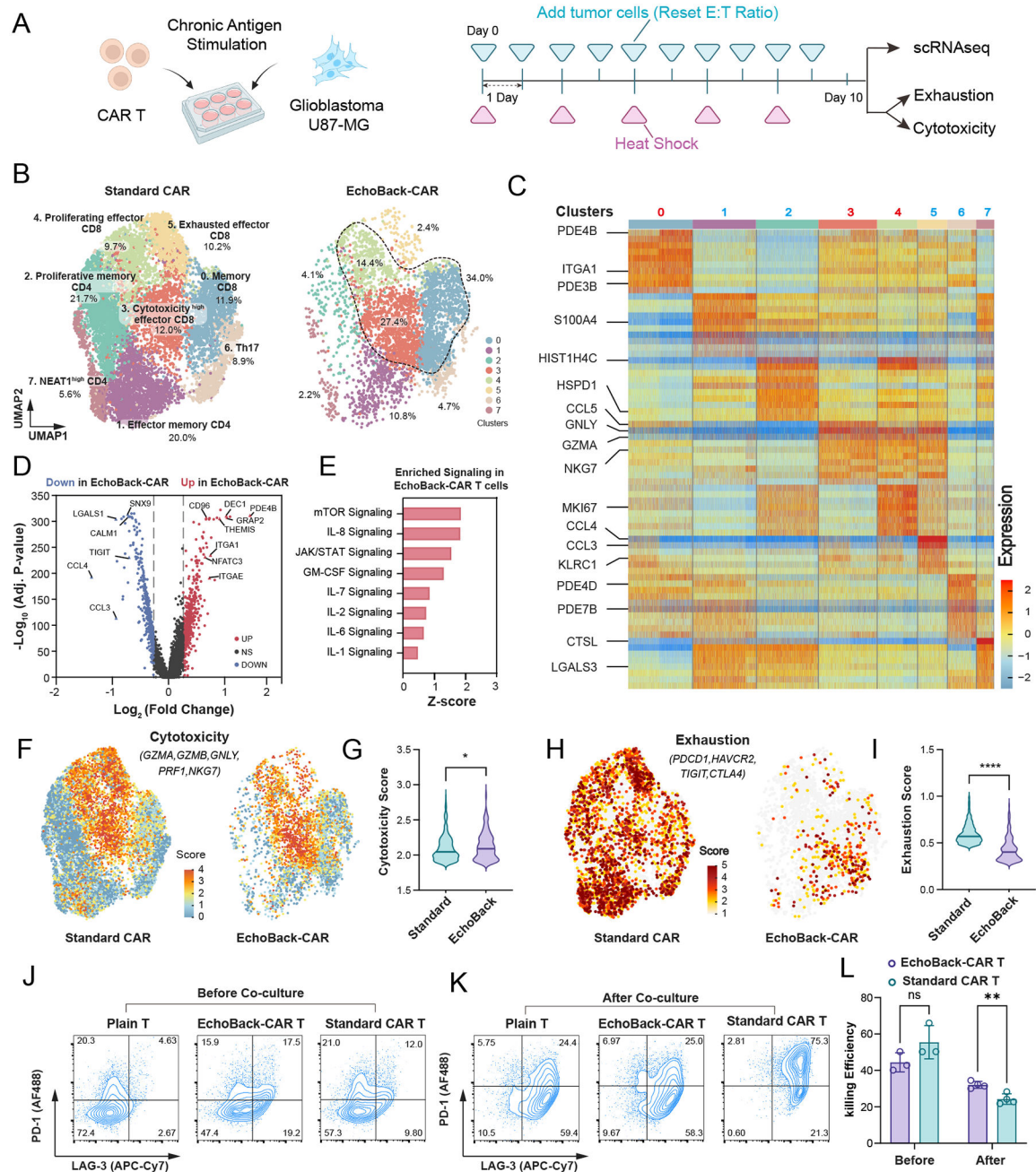


Figure 5. scRNAseq reveals the unique signaling network and functional features in EchoBack-CAR T cells.

A. Scheme of the repeated tumor stimulation assay. CAR-T cells were rechallenged with U-87 MG tumor cells at a 1:2 E:T ratio every 24 hours for 9 days. The E:T ratio was reset to 1:2 during each round of tumor stimulation. Heat shock at 43°C for 15 minutes was conducted every 48 hr.

B. UMAP visualization of standard and EchoBack-CAR T cells based on their transcriptomic profiles obtained by scRNA-seq. CAR-T cells were co-cultured with tumor cells as described in Figure 5A and collected on day 9 for single-cell RNA sequencing. Eight distinct clusters were identified and annotated. Clusters enriched in EchoBack-CAR

T cells (0, resident memory CD8⁺ T cells; 3, high-cytotoxicity effector CD8⁺ T cells; 4, proliferating effector CD8⁺ T cells) were circled with a dotted line.

C, Heatmap showing differentially expressed feature genes (rows) across cell clusters identified in Figure 5B. The number at the top of the heatmap corresponds to different clusters. Clusters enriched in EchoBack-CAR T cells (clusters 0, 3, and 4) are highlighted in red.

D, Volcano plot of the differentially expressed genes (DEGs) in EchoBack-CAR versus standard CAR-T cells. Genes on the right (red) and left (blue) are upregulated and downregulated, respectively, in EchoBack-CAR T cells relative to standard CAR-T cells after chronic antigen stimulation.

E, IPA results highlight the signaling pathways that are enriched in EchoBack-CAR T cells.

F, UMAP projections colored according to cytotoxicity scores of individual cells, based on the expression of cytotoxicity-related genes (GZMA, GZMB, GNLY, PRF1, NKG7) in EchoBack and standard CAR-T cells.

G, Violin plot depicting the cytotoxicity scores of standard and EchoBack CAR T cells from clusters 0, 3, and 4. Two-tailed unpaired t-test, *P = 0.0471.

H, UMAP projections colored according to cytotoxicity scores of individual cells, based on the expression of exhaustion-related genes (PDCD1, HAVCR2, TGIGT, CTLA4) in EchoBack and standard CAR-T cells.

I, Violin plot depicting the exhaustion scores of standard and EchoBack CAR T cells from clusters 0, 3, and 4. Two-tailed unpaired t-test, ****P < 0.0001.

J-K. Flow cytometry analysis illustrating the expression levels of exhaustion markers PD-1 and LAG-3 on CAR-T cells before (J) and after (K) exposure to a repeated tumor stimulation assay.

L. Cytotoxicity of the CAR-T cells before and after co-culture assay at E:T ratio of 1:2. n=3 in each group. Multiple unpaired t-test. ns p= 0.142361, **p= 0.00399. Error bars, mean ± SD.

See also Figure S5 and Figure S6.

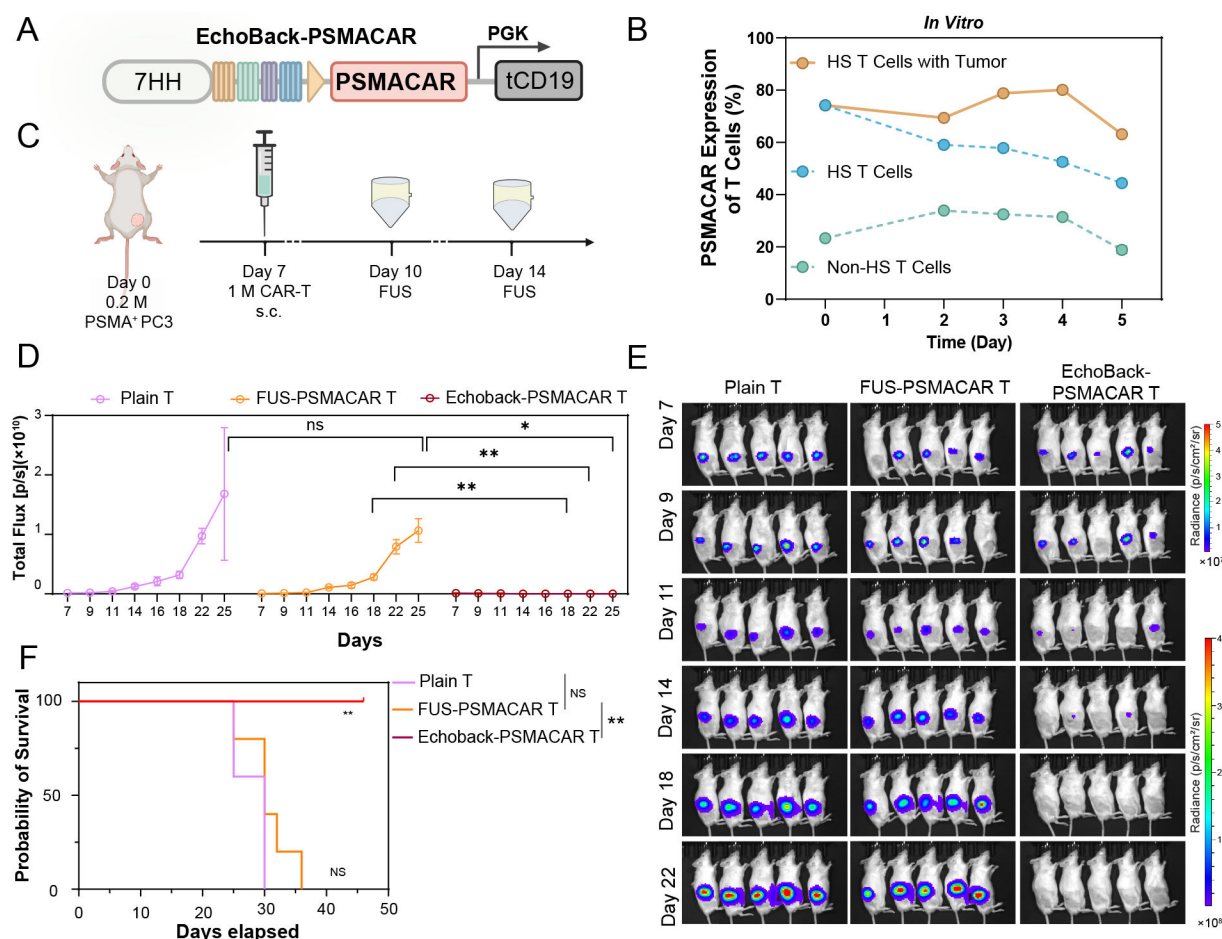


Figure 6. Examine EchoBack-PSMACAR targeting prostate tumor in vivo.

A. Schematics of the EchoBack-PSMACAR construct. The truncated CD19 (tCD19) is expressed as a marker for monitoring lentiviral transduction efficiency.

B. Expression of EchoBack-PSMACAR in human T cells under various conditions in vitro. CAR expression was quantified using antibody staining and flow cytometry. n=2 for each data point.

C. Experimental timeline for EchoBack-PSMACAR T therapy for the treatment of prostate cancer in mice.

D-E. Tumor growth in the prostate tumor model measured by bioluminescence imaging (D) and quantitative analysis of tumor volume over time post-treatment (E) with EchoBack-PSMACAR, FUS-PSMACAR, or plain T cells. The FUS-PSMACAR T and EchoBack-PSMACAR T group received the FUS stimulation. Two-way ANOVA, n=5 in each group, Day 18 **p=0.0093, Day 22 **p=0.0062, and Day 25 *p=0.0258. Error bars, mean $\pm$ SEM.

F. Survival curve of mice treated with plain T cells, EchoBack-PSMACAR T cells, or FUS-PSMACAR T cells. n=5 mice in each group. Log-rank (Mantel-Cox) test: ns p=0.2139, **p=0.0031.

See also Figure S7.

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Alexa Fluor 488 anti-human CD279 (PD-1) Antibody	Biolegend	Cat# 329936; RRID: AB_2563594
APC/Cyanine7 anti-human CD366 (Tim-3) [F38-2E2]	Biolegend	Cat# 345025; RRID: AB_2565716
APC/Cyanine7 anti-human CD223 (LAG-3) Antibody	Biolegend	Cat# 369219; RRID: AB_2910414
Brilliant Violet 421 anti-human Ganglioside GD2 Antibody	Biolegend	Cat# 357325; RRID: AB_2941536
Alexa Fluor 647 anti-human Ganglioside GD2 Antibody	Biolegend	Cat# 357317; RRID: AB_2800980
PE anti-human CD19 Antibody	Biolegend	Cat# 302208; RRID: AB_314238
Brilliant Violet 421™ anti-human CD4 Antibody	Biolegend	Cat# 317434; RRID: AB_2562134
APC/Cyanine7 anti-human CD8a Antibody	Biolegend	Cat# 301016; RRID: AB_314134
APC anti-human PSMA (FOLH1) Antibody	Biolegend	Cat# 342508; RRID: AB_10917056
Anti-CD8 alpha antibody [C8/468 + C8/144B]	Abcam	Cat# ab199016
Goat Anti-Mouse IgG H&L (HRP)	Abcam	Cat# ab205719
Alexa Fluor 647 AffiniPure F(ab') ₂ Fragment Goat Anti-Mouse IgG, F(ab') ₂ fragment specific	Jackson ImmunoResearch	Cat# 115-606-072
Bacterial and virus strains		
NEB 5-alpha Competent <i>E. coli</i> (High Efficiency)	New England Biolabs	Cat# C29871
ElectroMAX™ DH10B Cells	Thermo Fisher Scientific	Cat# 18290015
Biological samples		
Buffy coats from human healthy donors after donor deidentification	San Diego Blood Bank	N/A
Chemicals, peptides, and recombinant proteins		
ViaStain AOP1 staining solution	Revvity	Cat# CS2-0106
Recombinant human IL-2	PeproTech	Cat# 200-02
Lipofectamine 3000	Thermo Fisher Scientific	Cat# L3000008
Dulbecco's Modified Eagle Medium	Gibco	Cat# 11995115
Fetal bovine serum (FBS)	Gibco	Cat# 10438026
Penicillin–streptomycin (P/S)	Gibco	Cat# 15140122
Roswell Park Memorial Institute Medium (RPMI 1640)	Gibco	Cat# 22400105
X-VIVO 15	LONZA	Cat# 04-418Q
OptiMEM Reduced Serum Medium	Gibco	Cat# 31985
Trypsin-EDTA (0.05%)	Thermo Fisher Scientific	Cat# 25300054
Bladder Epithelial Cell Basal Medium	ATCC	Cat# PCS-420-032
Bladder Epithelial Growth Kit	ATCC	Cat# PCS-420-042
Human Recombinant EGF	Stemcell technologies	Cat# 78006.1
2-Mercaptoethanol	Gibco	Cat# 31350010
D-Luciferin, Potassium Salt (Proven and Published)	Gold Biotechnology	Cat# LUCK-100
Formalin solution, neutral buffered, 10%	Sigma-Aldrich	Cat# HT501128-4L
hydrogen peroxide, 3 wt.% solution in water, stabilized	Thermo Fisher Scientific	Cat# 426000010
Citrate Buffer, pH 6.0, 10×, Antigen Retriever	Sigma-Aldrich	Cat# C9999-100ML

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Blocker BSA	Thermo Fisher Scientific	Cat# 37525
Pierce DAB Substrate Kit	Thermo Fisher Scientific	Cat# 34002
StatLab Harris Hematoxylin Stain	Fisher Scientific	Cat# STL9016
Leica Biosystems MICROMOUNT MOUNTING MEDIA 1PT	Fisher Scientific	Cat# NC1109222
Ampicillin Sodium	Sigma-Aldrich	Cat# A0166
Diethylenetriaminepentaacetic acid gadolinium (III) dihydrogen salt hydrate (97%)	Sigma-Aldrich	Cat# 381667
DMSO	Sigma-Aldrich	Cat# D2650-100ML
Lenti-X Concentrator	Takara	Cat# 631232
LB broth	Fisher Scientific	Cat# BP14272
RetroNectin Recombinant Human Fibronectin Fragment	Takara	Cat# T100A
Critical commercial assays		
Dynabeads Human T-Expander CD3/CD28	Gibco	Cat# 11141D
ProFection Mammalian Transfection System	Promega	Cat# E1200
ONE-Glo Luciferase Assay System	Promega	Cat# E6110
Passive Lysis 5X Buffer	Promega	Cat# E1941
Pan T Cell Isolation Kit	Miltenyi Biotec	Cat# 130-096-535
Corning LSM (Lymphocyte Separation Medium)	Corning	Cat# 25-072-CV
HiSpeed Plasmid Maxi Kit	Qiagen	Cat# 12663
ZymoPURE Plasmid Miniprep Kit	ZymoResearch	Cat# D4211
Zymoclean Gel DNA Recovery Kit	ZymoResearch	Cat# D4008
Quick-DNA Microprep Kit	ZymoResearch	Cat# D3020
Deposited data		
Raw and processed sequencing data	This paper	GSE289368
Experimental models: Cell lines		
Lenti-X 293T cells	Clontech Laboratories	Cat# 632180
Jurkat, Clone E6-1	ATCC	Cat# TIB-152
HIEC-6	ATCC	Cat# CRL-3266
BdEC	ATCC	Cat# PCS-420-010
293T	ATCC	Cat# CRL-3216
U-87 MG	ATCC	Cat# HTB-14
Experimental models: Organisms/strains		
NSG mice (NOD.Cg-Prkdc ^{scid} Il2rgt ^{m1Wjl} /SzJ)	Jackson Laboratory	Cat# 005557
Oligonucleotides		
Primers for Library Screening	IDT	See Table S1
Recombinant DNA		
pLVX-lacZ-SFFV-mCherry	This work	N/A
pHR-PGK-hGD2CAR-P2A-mCherry	This work	N/A
pHR-EchoBack-hGD2CAR-PGK-tCD19	This work	N/A
pHR-EchoBack-PSMACAR-PGK- mCherry	This work	N/A

REAGENT or RESOURCE	SOURCE	IDENTIFIER
pHR-Hsp-PSMACAR-PGK-tCD19	This work	N/A
pHR-PGK-PSMACAR	This work	N/A
pHR-PGK-Firefly Luciferase-P2A-EGFP	This work	N/A
pMD2.G	Addgene	Cat# 12259
PsPAX2	Addgene	Cat# 12260
Software and algorithms		
Living Image v4.4	PerkinElmer	Perkinelmer.com
FlowJo 10.7.1	BD bioscience	Flowjo.com
Sanpgene	Snapgene	www.snapgene.com
Graphpad Prism 9	GraphPad Software	graphpad.com/scientific-software/prism
R Studio	R Studio	Rstudio.com
IncuCyte 2022A	Sartorius	www.sartorius.com/en/products/live-cell-imaging-analysis/live-cell-analysis-software
Adobe illustrator	Adobe	www.adobe.com
ZEISS ZEN lite	ZEISS	www.zeiss.com/microscopy/us/products/software/zeiss-zen-lite.html
Other		
Gibson assembly master mix	New England Biolabs	Cat# E2611L
Q5 DNA Polymerase	New England Biolabs	Cat# M0491
0.45 μ m PES filters	Genesee Scientific	Cat# 25-246
μ -Plate 96 Well Square plates	ibidi	Cat# 89621
PZT material	Del Piezo Specialties	Cat# DL-47
Aquasonic 100 Ultrasound transmission gel	Aquasonic gell	Cat# PLI 01-08
Mylar Thin-film	Chemplex Industries	Cat# 100
Thermometers	Omega	Cat# HH806AU
MT-29/2HT Needle Microprobe	Physitemp Instruments	MT-29/2HT
Function generator	Sanford Research System	Cat# SG386
50 dB power amplifier	E&I	Cat# 325LA